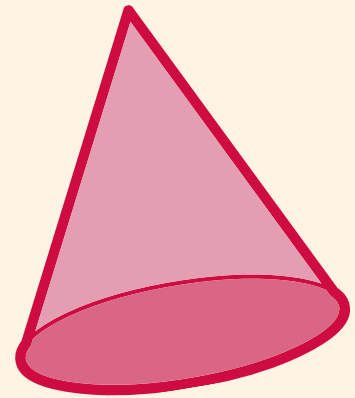
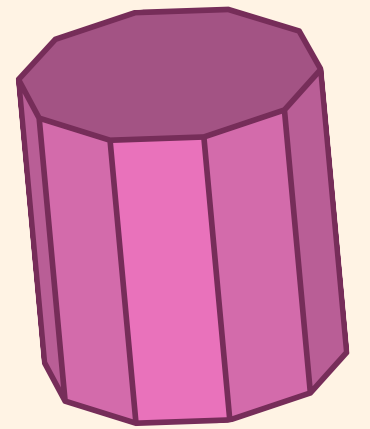
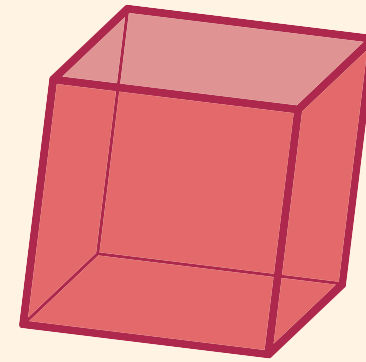
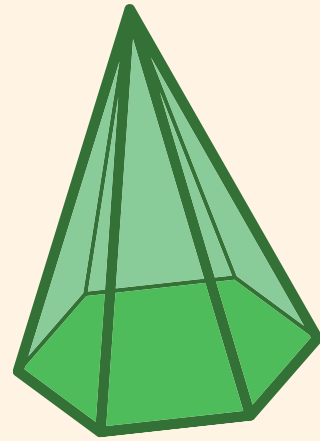
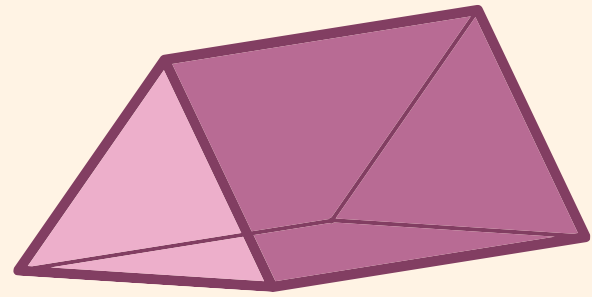
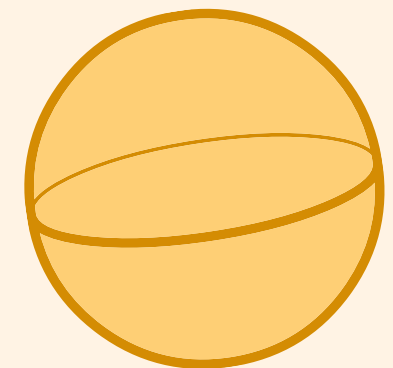
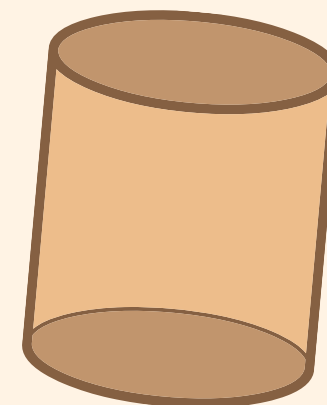
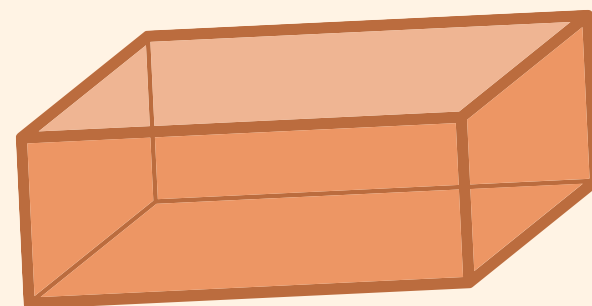
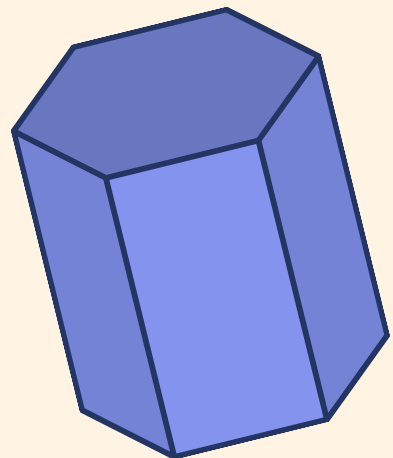
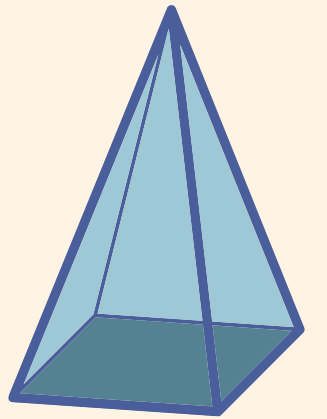
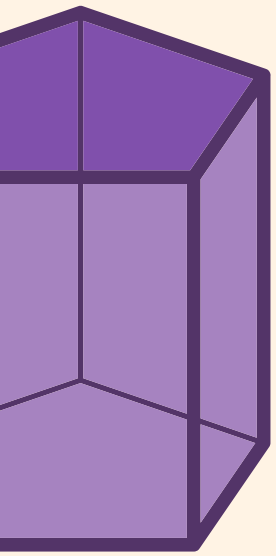




3D Shapes

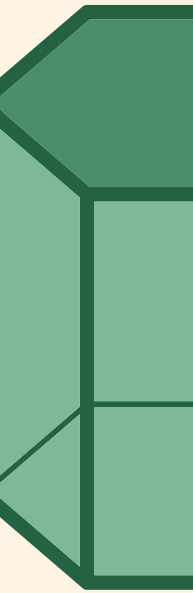


Learning Objective:



Use mathematical
names for 3D
shapes

Know the
properties of 3D
shapes

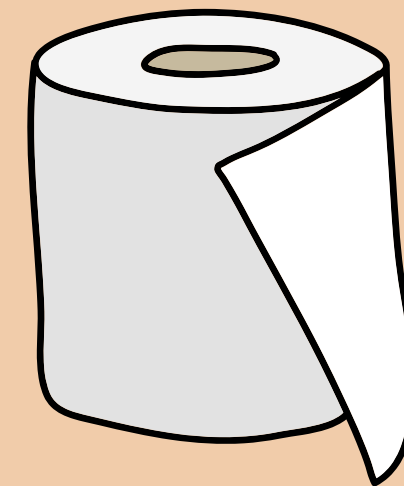


What are 3D shapes?

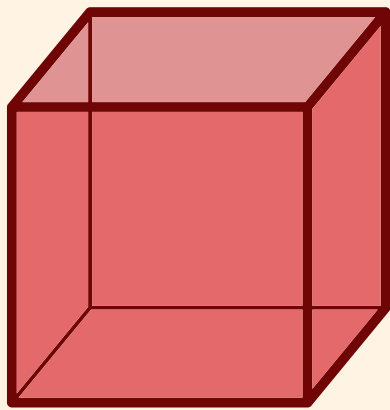
3D shapes, or three-dimensional shapes, are objects that have length, width, and height.

We will explore the mathematical names of 3D shapes and their properties.

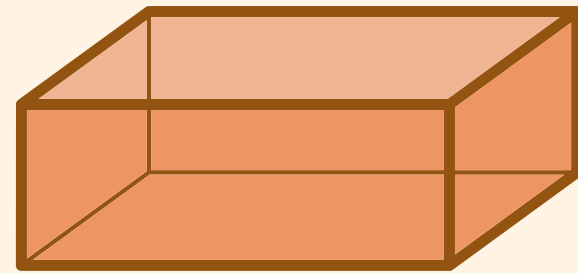
Real life objects:



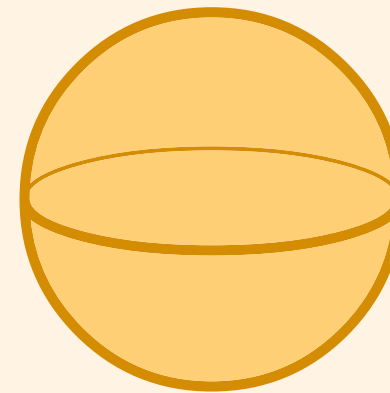
Common 3D shapes



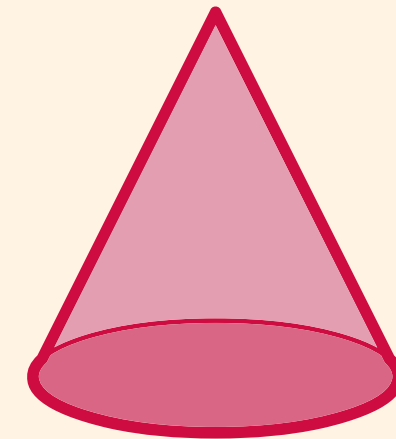
Cube



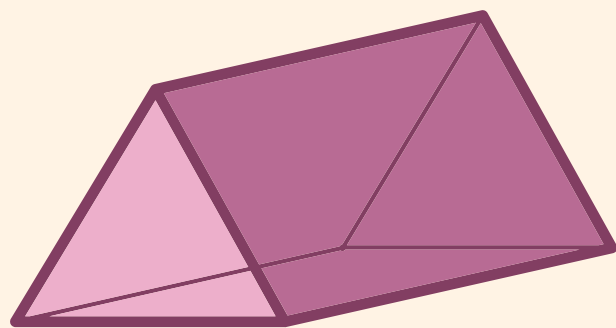
Cuboid



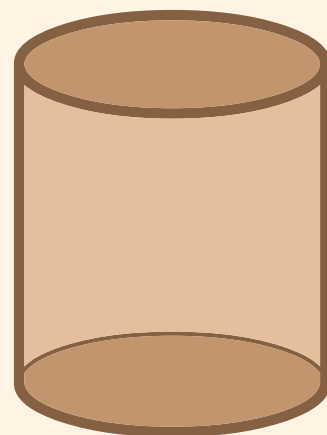
Sphere



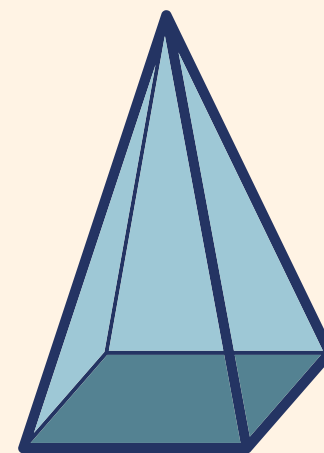
Cone



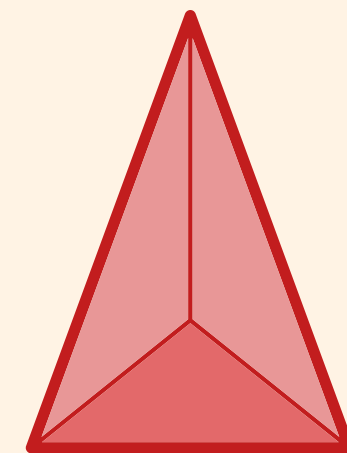
Triangular
Prism



Cylinder



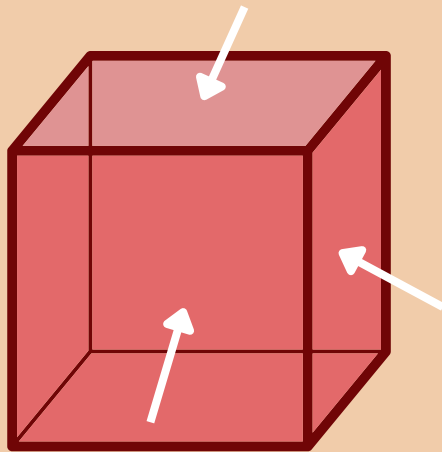
Square based
pyramid



Tetrahedron

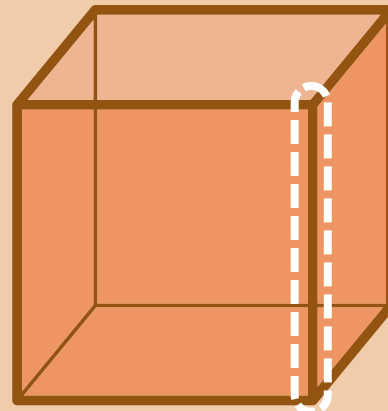
Properties of 3D shape

Faces



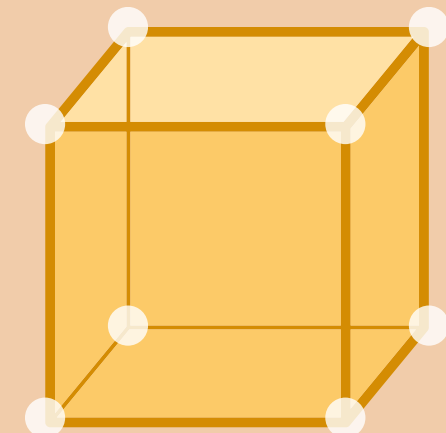
Faces are flat or curved surfaces on a 3D shape

Edges



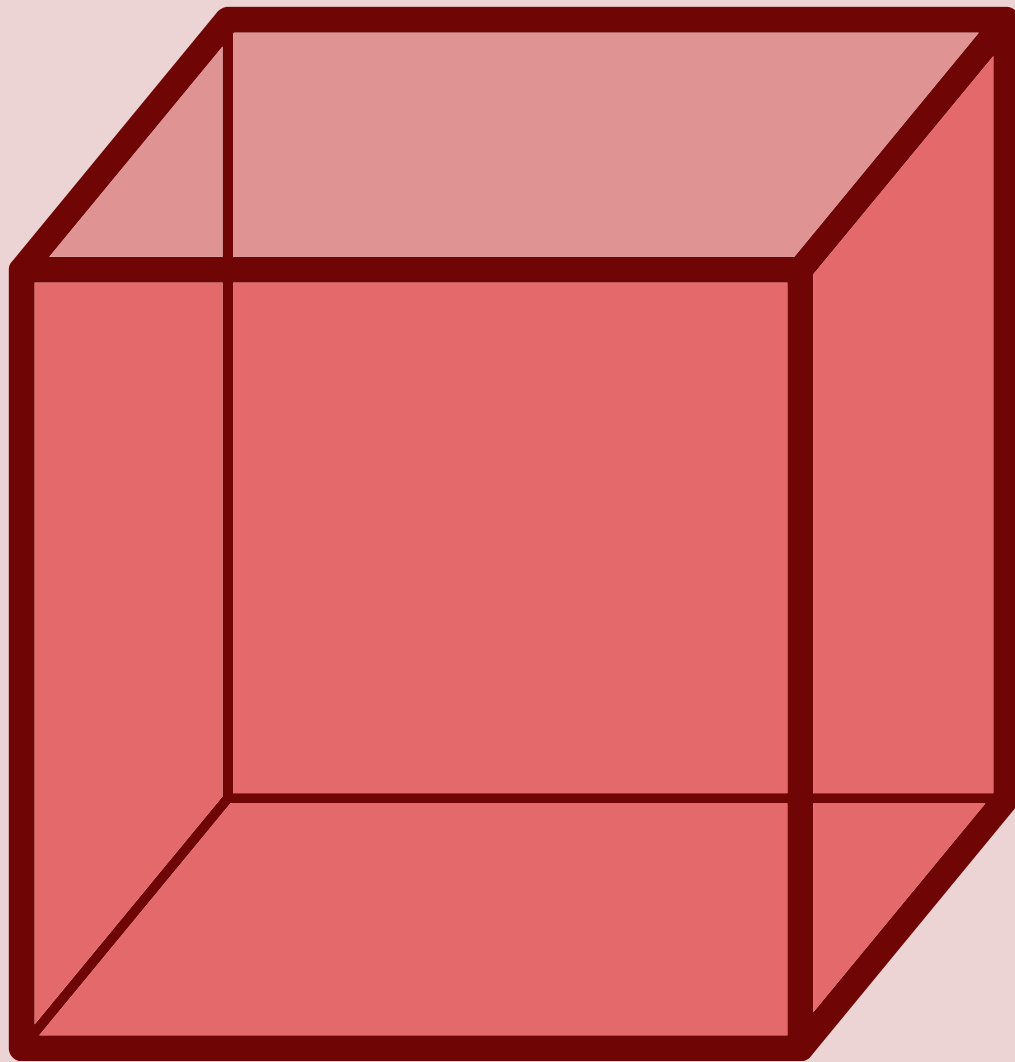
An edge is where two faces meet

Vertices



Vertices/corners are where two or more edges meet

Cube



Faces

6

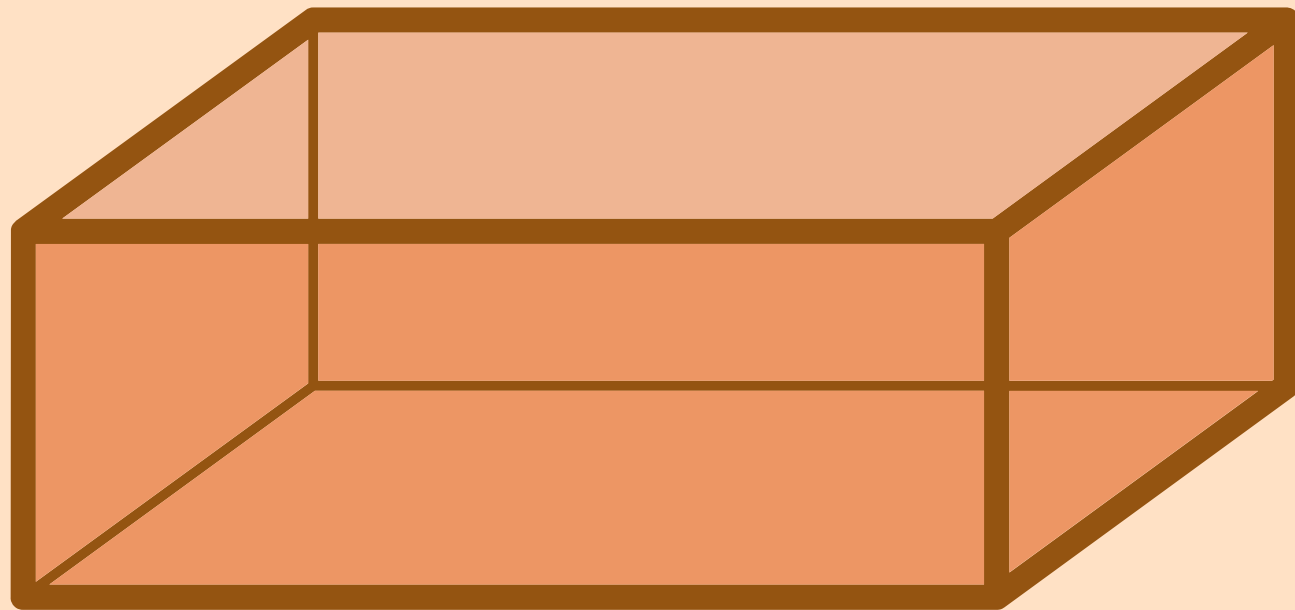
Edges

12

Vertices

8

Cuboid



Faces

6

Edges

12

Vertices

8

Sphere



Faces

1

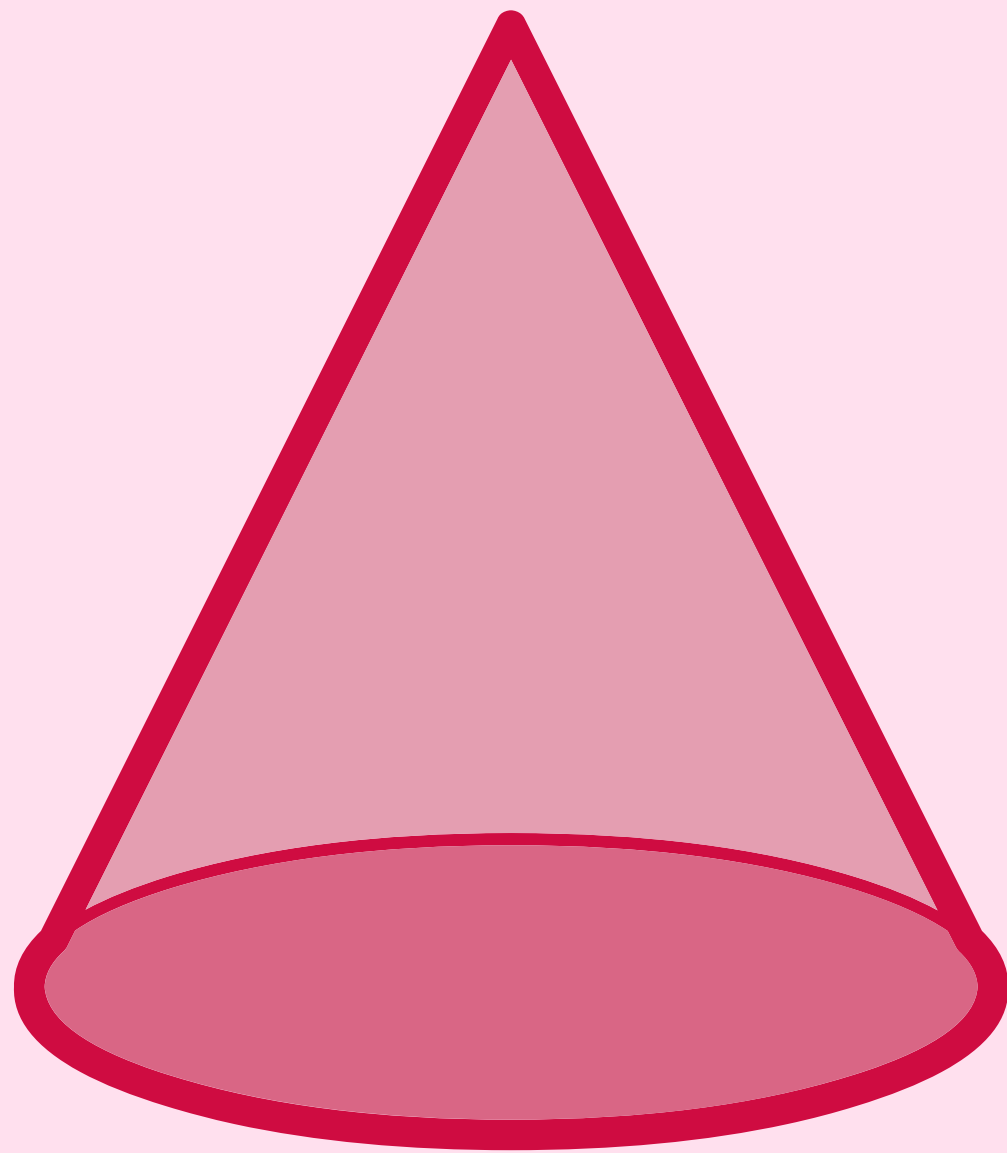
Edges

0

Vertices

0

Cone



Faces

2

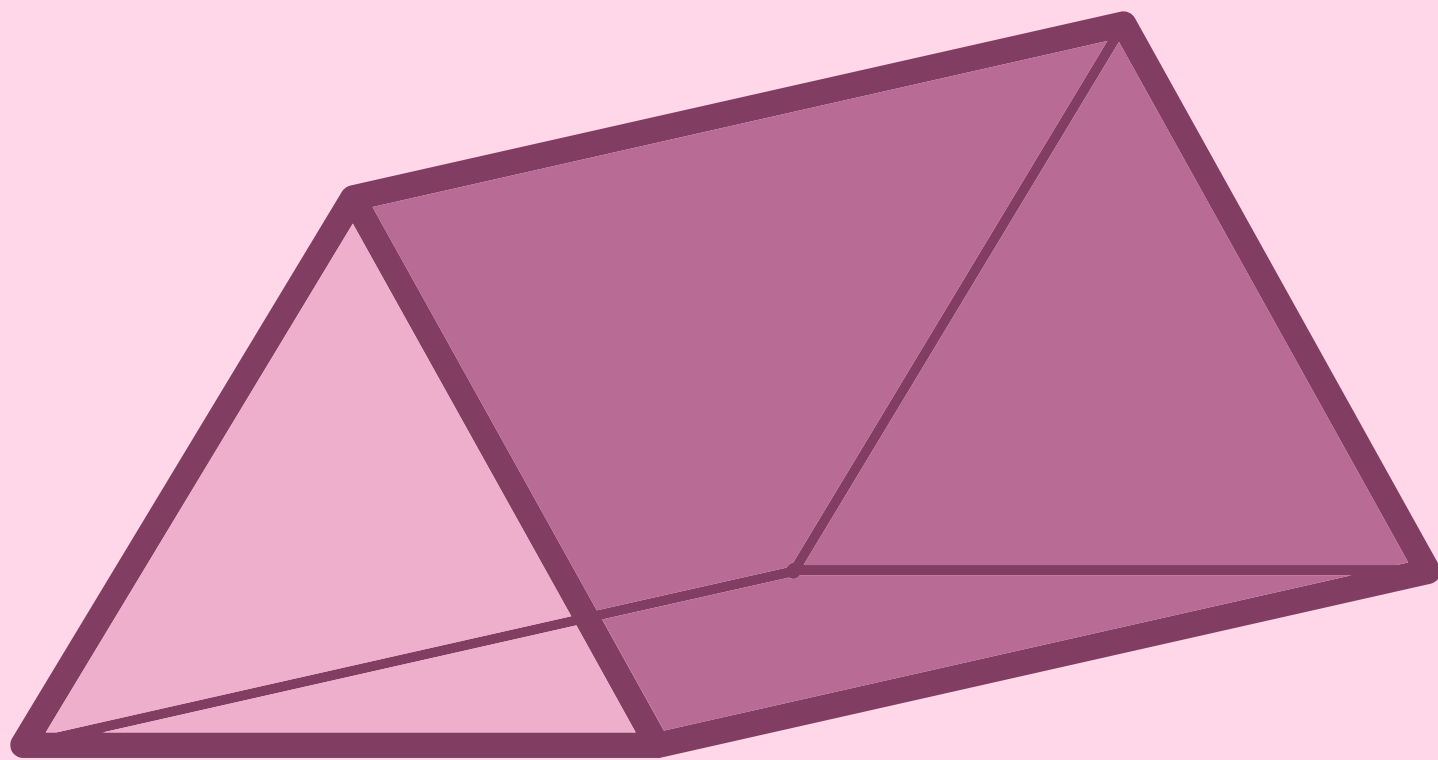
Edges

1

Vertices

1

Triangular Prism



Faces

5

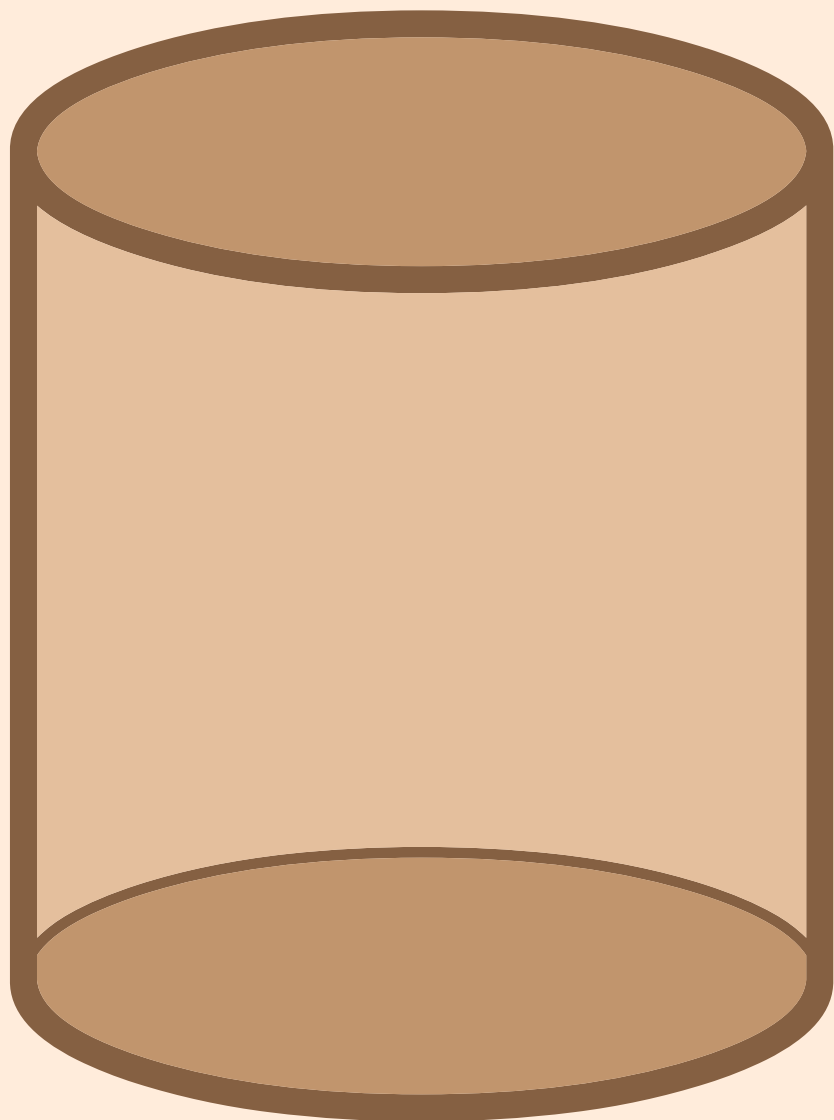
Edges

9

Vertices

6

Cylinder



Faces

3

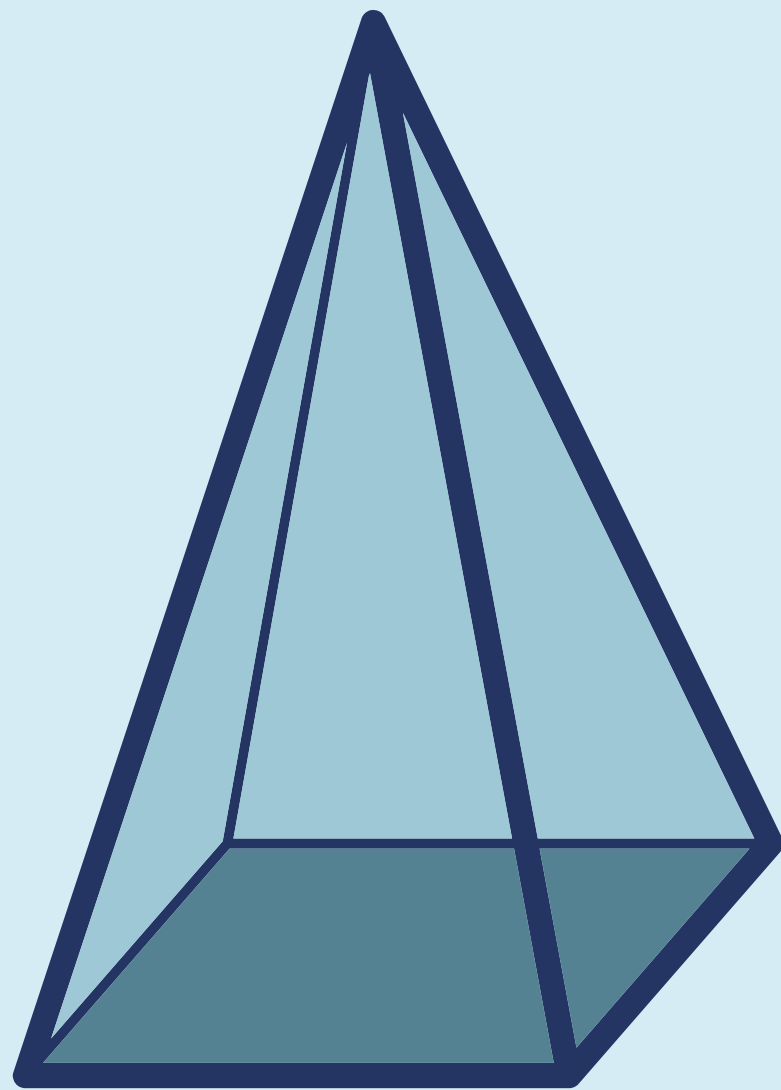
Edges

2

Vertices

0

Square-based Pyramid



Faces

5

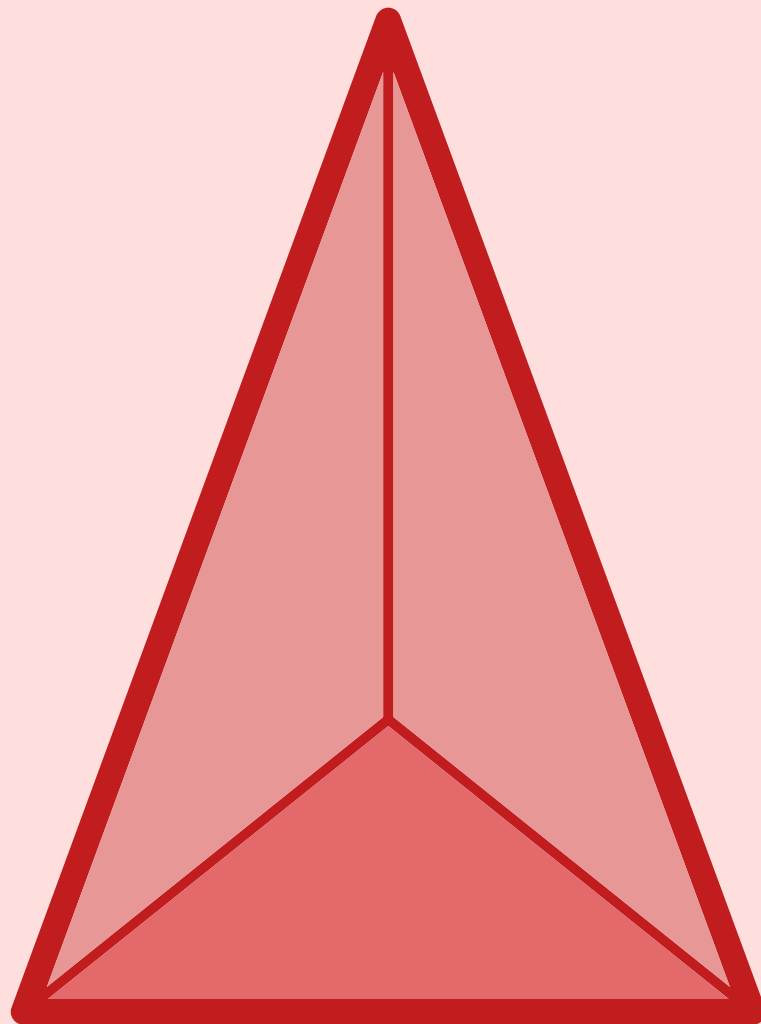
Edges

8

Vertices

5

Tetrahedron



Faces

4

Edges

6

Vertices

4

**Thank you
for listening**

Nets of 3D

SHAPES

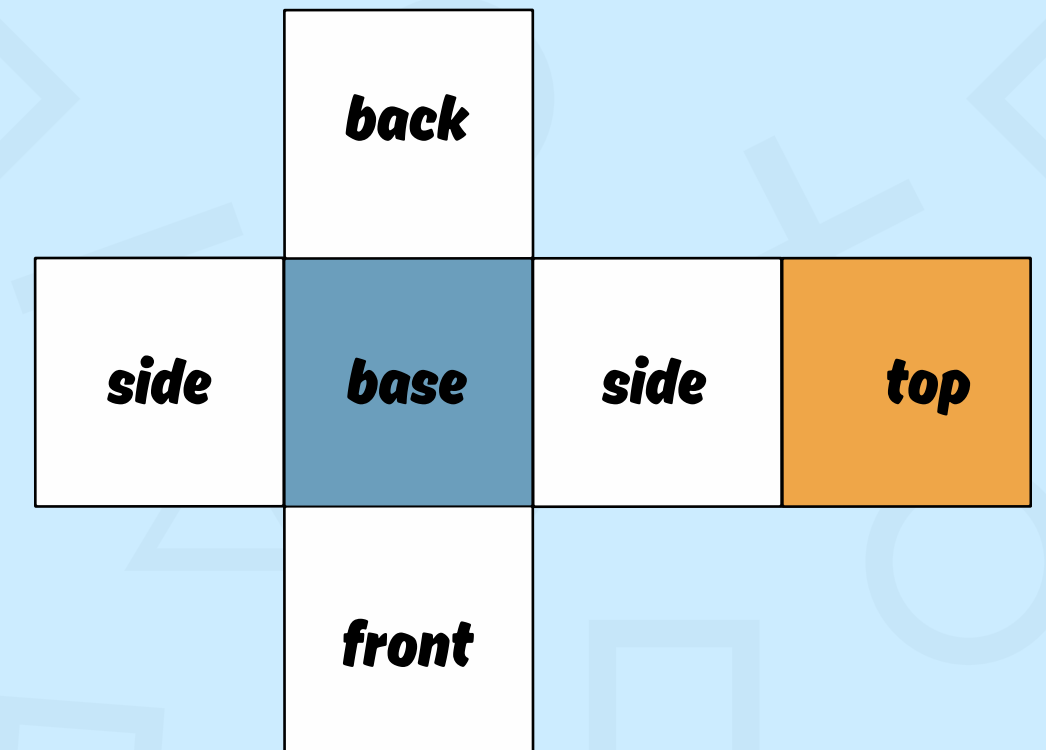
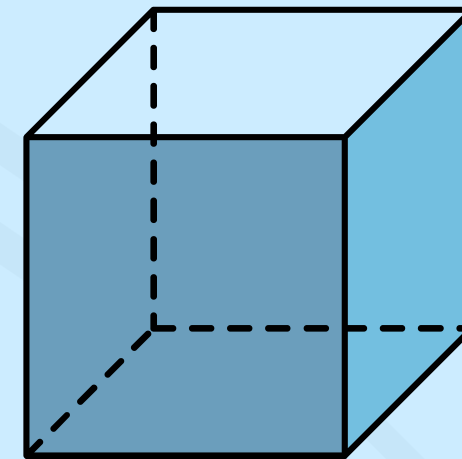


What is a net?

A net is a three-dimensional shape that has been flattened out and opened up. It may be folded to form a 3D shape and indicates the number and type of faces a 3D shape contains.

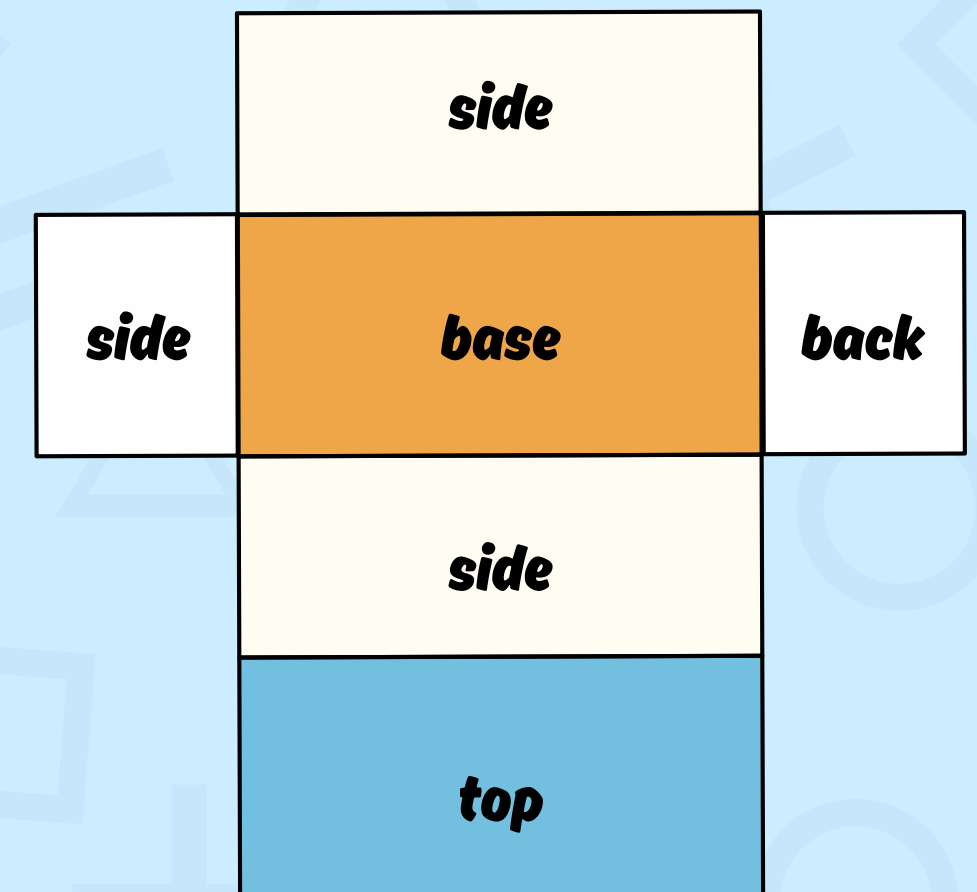
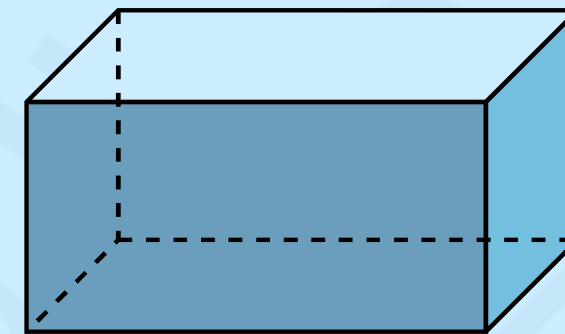
Net of a Cube

- A cube has 6 faces.
- A cube consists of 6 square faces that are connected together.
- When a cube is unfolded it will give 6 squares.
- All faces are congruent and perpendicular to each other.
- There are 11 nets in a cube.



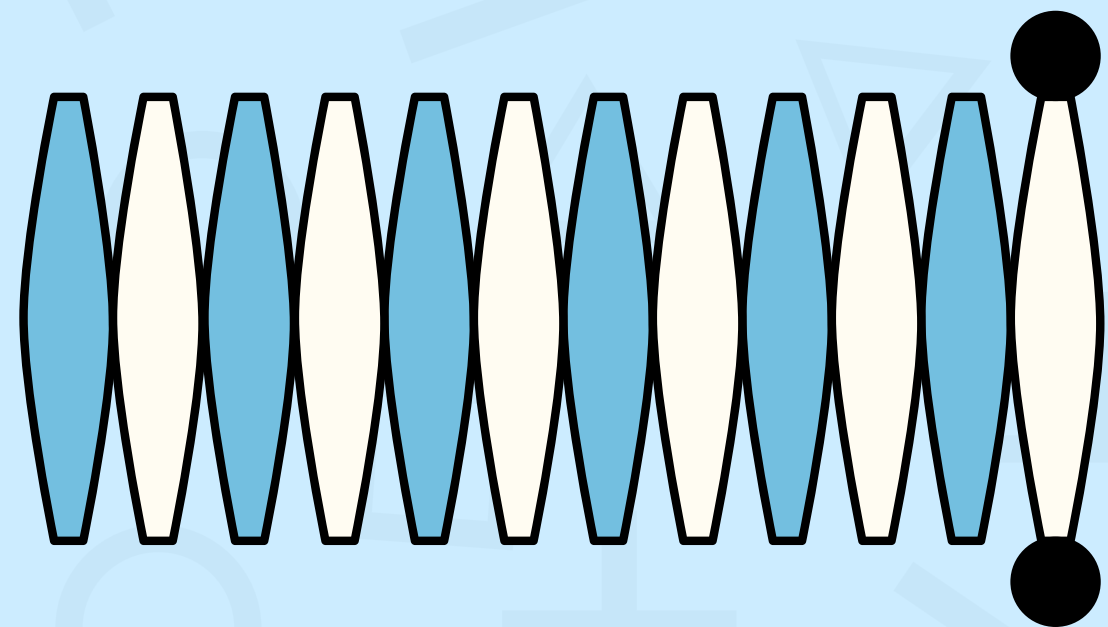
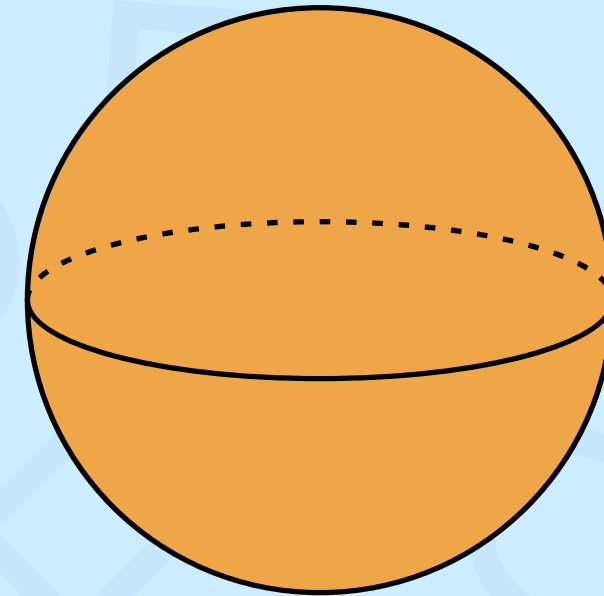
Net of a Cuboid

- A cuboid has 6 faces and all of these faces are rectangles
- If a cuboid is unfolded, it will give 6 rectangles.
- Opposite faces are congruent and parallel to each other.
- There are 54 different nets of the cuboid.



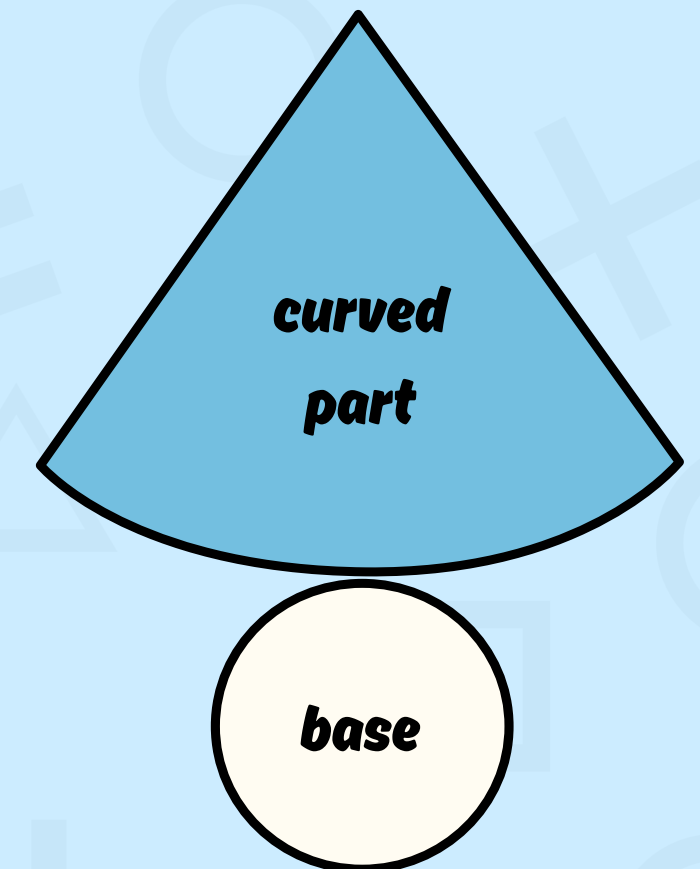
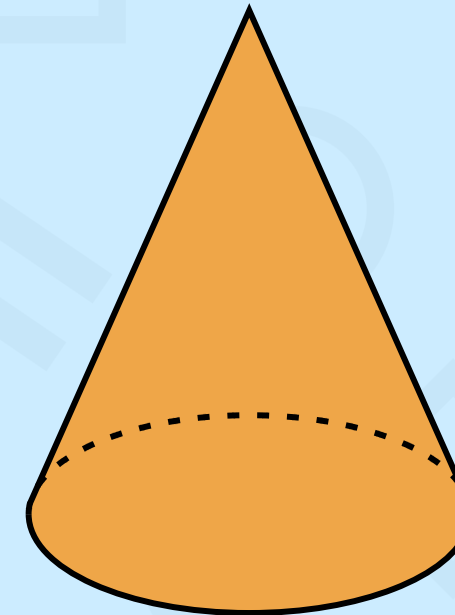
Net of a Sphere

- A sphere is a perfectly symmetrical 3d shape that has no flat surface.
- It has all point equidistant from the center.
- When a sphere is unfolded, it will give a continuous curve.



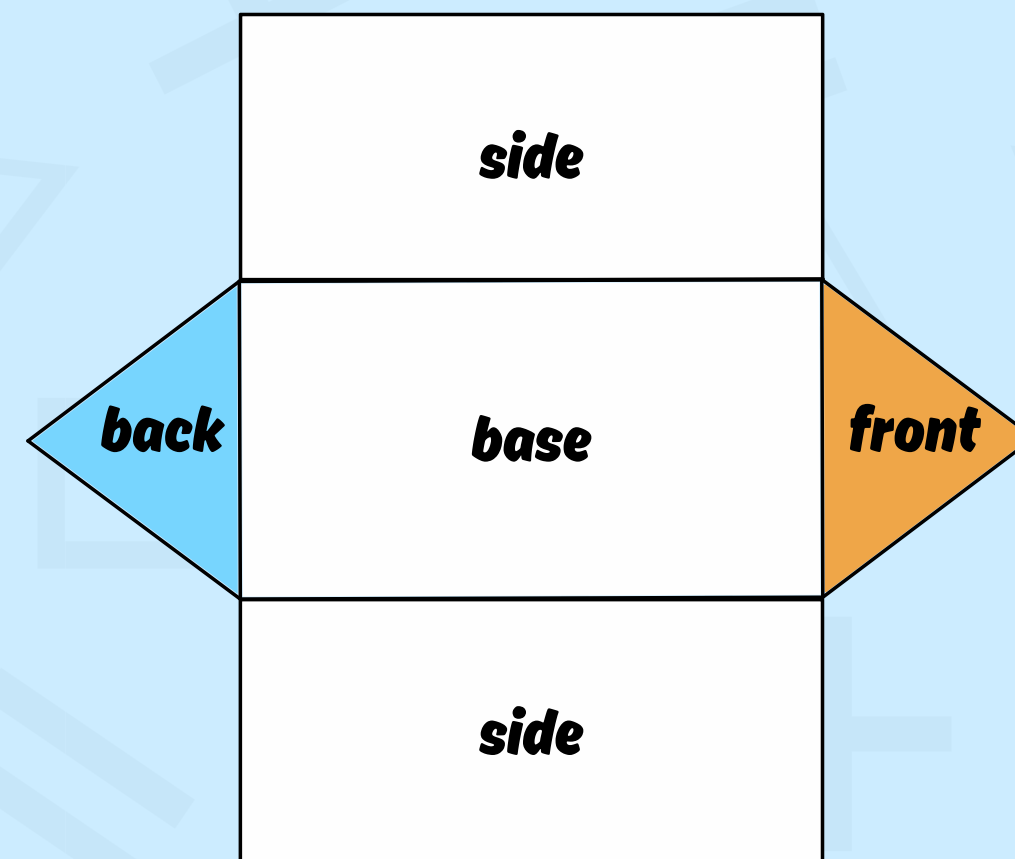
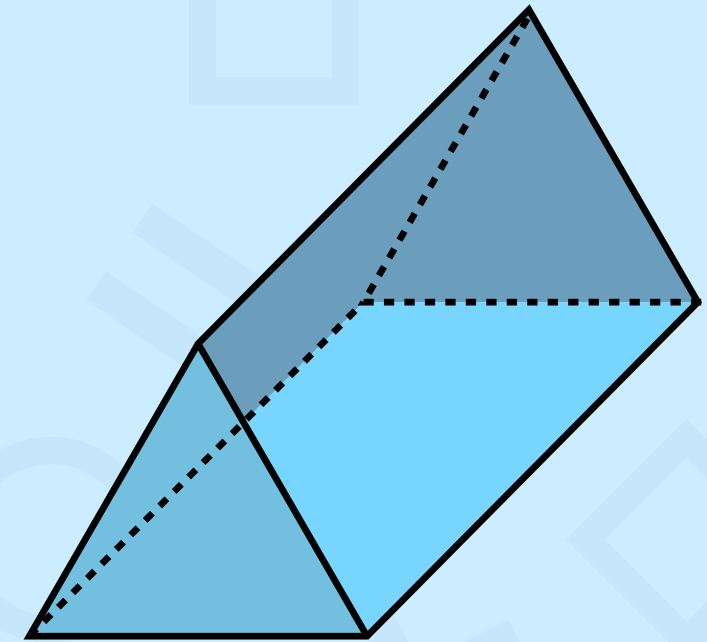
Net of a Cone

- The cone has only 2 faces.
- Out of the 2 faces, 1 is a curved surface and the other one is a flat face.
- A net of cones has 2 parts: One is the circle that represents the flat part. The second one represents the curved part.
- When it is unfolded, it will give 1 circle and 1 sector that tapers to a single point called apex.



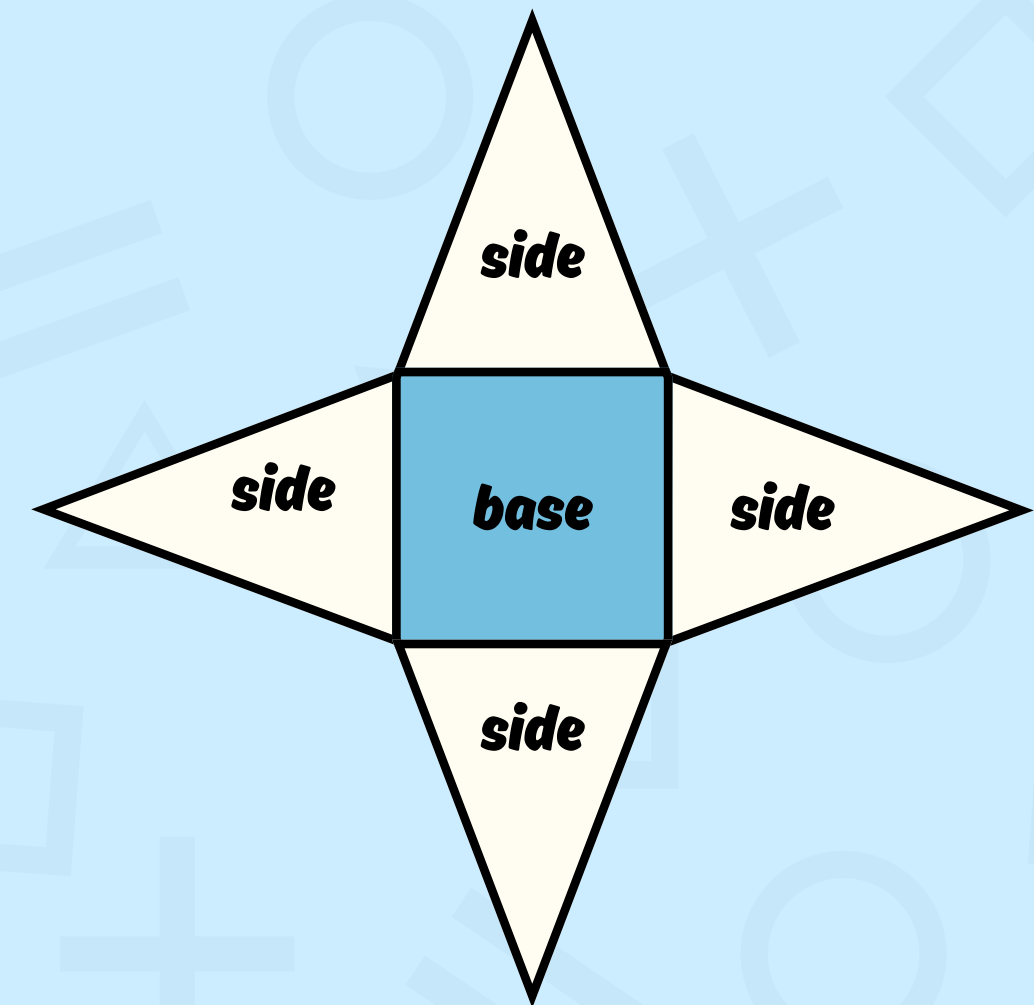
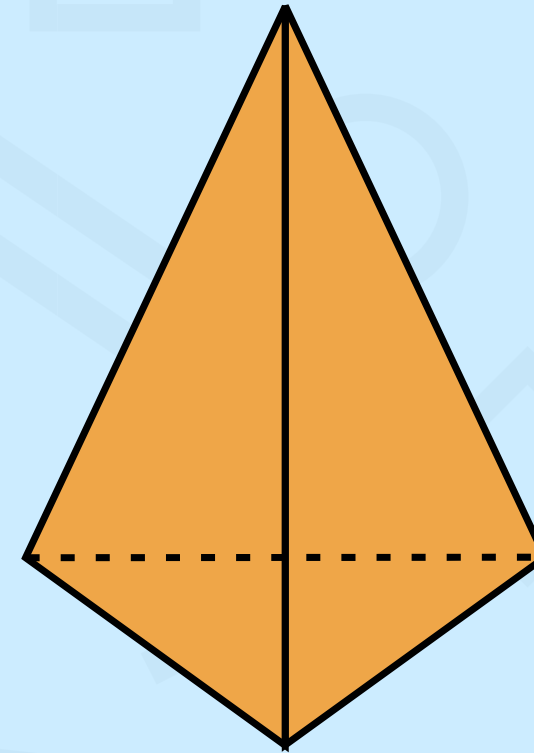
Net of a Prism

- A prism has 2 identical, congruent polygons that form from the top and bottom of the prism.
- The triangular prism has 2 triangular bases and 3 rectangular lateral faces connecting the corresponding edges of the triangles.



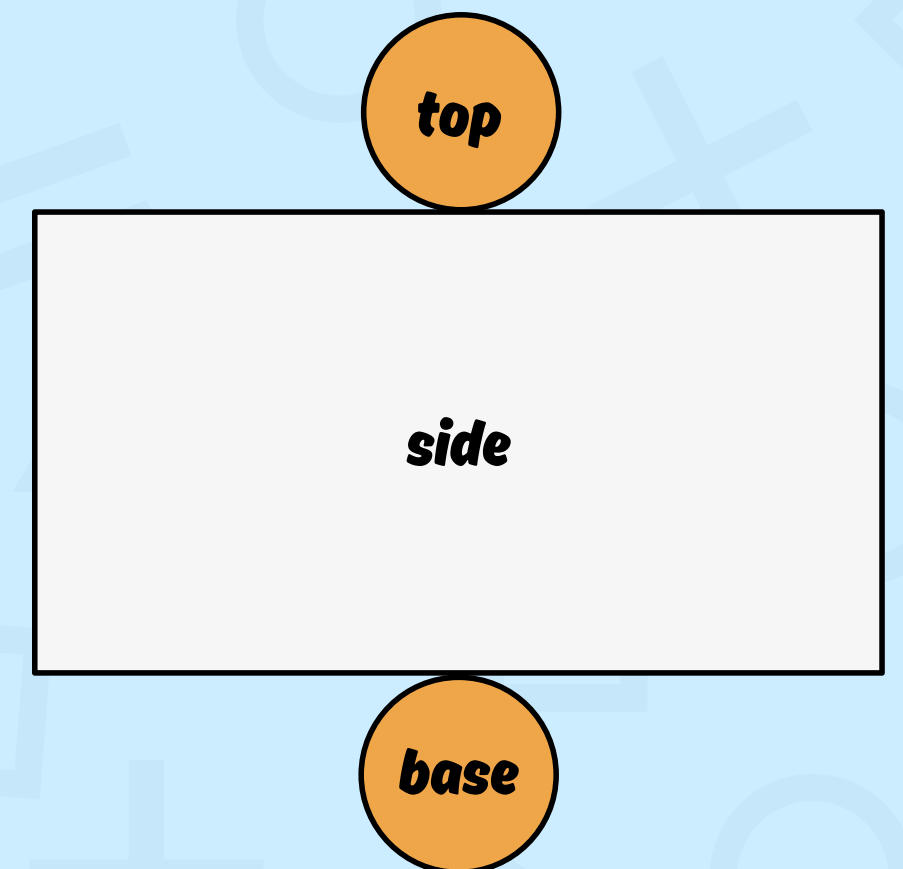
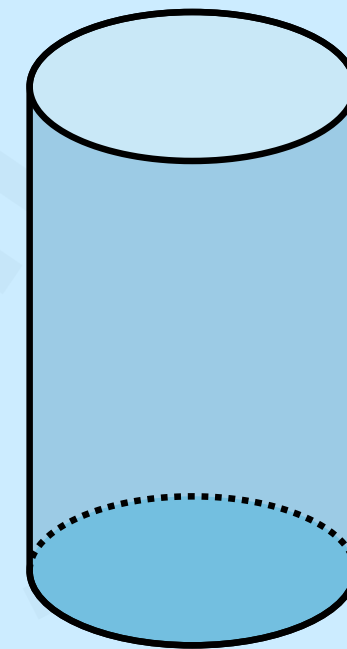
Net of a Pyramid

- A pyramid has 5 faces.
- These faces include 1 square and 4 triangular faces.
- When a pyramid is unfolded it will give 1 square and 4 triangles.



Net of a Cylinder

- A cylinder is made up of 3 faces.
- Two are circular bases and one is a curved surface.
- Two of the faces are circular and one is rectangular.
- When a cylinder is unfolded, it will give 2 circles and 1 rectangle.



Nets of 3D

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Thank you for today!